

Unraveling hidden Raman modes in α -MnTe: Effects of long-range dipole-dipole interactions on phonon bands

Susmita Jana^{1,2,*}, Rajib Sarkar³, Dhavala Suri³, and B.R.K Nanda^{1,2†}

¹Condensed Matter Theory and Computational Lab,

Department of Physics, IIT Madras, Chennai, 600036, Tamil Nadu, India

²Center for Atomistic Modelling and Materials Design,
IIT Madras, Chennai, 600036, Tamil Nadu, India and

³Centre for Nanoscience and Engineering, Indian Institute of Science, Bengaluru, Karnataka 560012, India

In this study, we address the seemingly open question of disparity between the experimentally and theoretically observed Raman spectra for the altermagnetic α -MnTe. We establish that the consideration of long range (LR) dipole-dipole interaction is crucial to accurately estimate the phonon band structure and thereby capturing each of the experimentally observed Raman signatures. The LR correction brings about substantial changes to the high-frequency optical modes and their group-theoretical representations. Specifically, for the highly debated Raman peak around 175 cm^{-1} , alongside the Raman silent B_{1u} mode, the LR corrections introduce Raman-active longitudinal optical A' and A_1 , and transverse optical A_1 modes. Overall, the calculated Raman spectra suggest that the intensity of these modes is weak in the pristine system, while certain prescribed distortions can increase their intensity. The Raman peak around 140 cm^{-1} , as observed experimentally, is identified to be emerging from a higher-order Raman process due to zone off-center contributions, while the other debated 120 cm^{-1} peak has both the zone center and off-center contributions.

Over the last few years the insulating α -MnTe has been intensely investigated as it exhibits non-relativistic spin splitting of the bands, a characteristic feature of altermagnetism[1], and finite anomalous Hall conductivity[2, 3]. Alongside the electronic and magnetic structure[4, 5], experimental Raman characterization, one of the fundamental sample characterization tools, has opened up a new fascinating research problem involving this compound. Broadly, the experimental results are segregated into two categories, one with the presence of Raman peak at 175 cm^{-1} [6–9] and the others which do not observe such peaks[10–12].

The MnTe films that are grown on InP (111) substrate, as reported by Zhang *et. al.*[7] show Raman peaks at 122.5 cm^{-1} , 140.6 cm^{-1} , 175.7 cm^{-1} , 268.8 cm^{-1} , and 345.8 cm^{-1} . The peak at 175.7 cm^{-1} is assigned to the E_{2g} mode. There are some other experimental studies which have not reported the presence of Raman peak at around 175 cm^{-1} . For example, two of the earlier Raman measurements[10, 13] on μm thick MnTe film, carried out at low temperature ($< 30\text{K}$) do not exhibit excitation at 175 cm^{-1} . However, they report Raman excitation in the range $195 - 215\text{ cm}^{-1}$. A very recent experimental work reveal that for the MnTe films grown on GaAs (111) substrate, show Raman peaks at 90 cm^{-1} , 120 cm^{-1} , and 140 cm^{-1} [11]. Interestingly, the air exposed samples in this study did not exhibit the 175 cm^{-1} peak even though one would expect oxidation and Te excess surface in them. In a recent experimental study on MnTe film MBE grown on Si(111) substrate, we notice a mixed observation. The pristine spot also showed the Raman intensity peaks around $90\text{--}105\text{ cm}^{-1}$, 120 cm^{-1} ,

and 140 cm^{-1} . The spot burned with laser irradiation showed these peaks and in addition also showed the peak at 175 cm^{-1} . [12] By carrying out a thorough analysis, we identified the mode associated with this peak as a signature of extrinsic MnTe_2 impurity or of distorted MnTe, rather than an intrinsic phonon or plasmon feature. It may be noted that MnTe_2 has a well established A_g peak at 175 cm^{-1} [14].

In the case of bulk MnTe single crystal, grown using chemical vapour transfer method, polarization dependent Raman measurements show peaks at 120 cm^{-1} , 140 cm^{-1} , 175 cm^{-1} , and 270 cm^{-1} [8]. In the pristine system, the Γ -centered phonon mode around the highest optical frequency, 175 cm^{-1} is Raman silent, and the authors propose that a probable distortion, reducing the D_{6h} to D_{3h} point group symmetry, converts the B_{1u} mode to A_1' , and thereby making it Raman active. Similarly, another study[9] involving bulk single crystal MnTe also reports the Raman peak at 175 cm^{-1} . However, the authors here assigned the peak to the possible presence of residual MnTe_2 as they did not observe any symmetry lowering in the pristine system. Differing with the above observation, a report on bulk MnTe assigned 175 cm^{-1} Raman peak in α -MnTe to E_{2g} mode [6] (see the supplementary material of this paper).

Concurrent to the experimental studies, several first-principles calculations are carried out to obtain the phonon modes and thereby the Raman signatures in MnTe. Thapa *et. al.* ruled out the possibility of E_{2g} mode to be responsible for the Raman excitation at 175 cm^{-1} . Instead, they proposed it as an outcome of a plasmonic (electronic excitations) origin, generated due to self hole doping. Specific to 175 cm^{-1} peak, Wu *et. al.*[8] proposed that, a plausible symmetry lowering distortion in highly symmetric MnTe can result in a leakage of 175 cm^{-1} mode. The theoretical calculations of Raman ac-

* jsusmita.research@gmail.com

† nandab@iitm.ac.in

tive modes reported in [11] did not identify the concerned feature while accounting the in-plane tensile strain induced by the GaAs substrate and its temperature dependence. A systematic DFT+U study by Uykur *et al.* [9] in MnTe with different Hubbard U values, and different exchange-correlation functionals reveals the presence of two closely spaced Raman active modes in the range $\sim 86\text{-}102\text{ cm}^{-1}$ and no other Raman active modes. Interestingly, the experimental study in this work reports the existence of two other strong modes at 120 cm^{-1} and 140 cm^{-1} . The later are believed to be non Γ -centered modes which are activated by higher order Raman processes. This is a feature which has not been substantiated by any other works.

So far, the widely reported Raman peaks have not been clearly assigned to the zone-center optical phonons of the α -MnTe. Therefore, a complete understanding of the phonon band structure of MnTe and its associated Raman active modes are far from complete. The contradictions among the varied reports as mentioned above has only increased the curiosity to resolve this seemingly open question. In this work, we present the phonon band structure of MnTe, and identify all possible Raman active modes and discuss the possible reasons for the missing ones.

The harmonic forces required to analyze the vibrational character of MnTe were calculated within density functional theory (DFT) as implemented in VASP. The interatomic forces were calculated using the PAW method with the PBE exchange-correlation functional of GGA. The crystal structure was fully relaxed with respect to both lattice parameters and atomic positions while retaining the A type Néel order of this altermagnet. To account the effect of correlations, DFT+U method is employed within the rotationally invariant Dudarev approach[15]. A symmetric $2\times 2\times 2$ supercell is used to incorporate the vibrations of atoms at the primitive cell boundary. After calculating a set of converged forces using density-functional perturbation theory (DFPT), the dynamical matrices were obtained on a $9\times 9\times 3$ k -mesh and, the same set of convergence parameters were used as in the ground-state calculations. The resulting phonon eigenvalues and eigenvectors were subsequently calculated as a solution of the dynamical matrix using Phonopy[16].

The DFT+U ($U = 4\text{ eV}$) calculated phonon band structure and atom-projected phonon density of states of MnTe are shown in Fig. 1(a). This band structure is in agreement with the ones reported in the literature[11]. The band structure is obtained by unfolding the phonon calculations performed using a $2\times 2\times 2$ supercell onto the Brillouin zone (BZ) of the primitive unit cell. For comparison, the folded supercell phonon band structure in the region of interest ($150\text{--}200\text{ cm}^{-1}$) is shown in Fig. 1(b), where we observe that there are six high-frequency solutions of the dynamical matrix lying at Γ

The highest optical mode (1), at 185.09 cm^{-1} , corresponds to a symmetric transverse optical (S-TO) vibra-

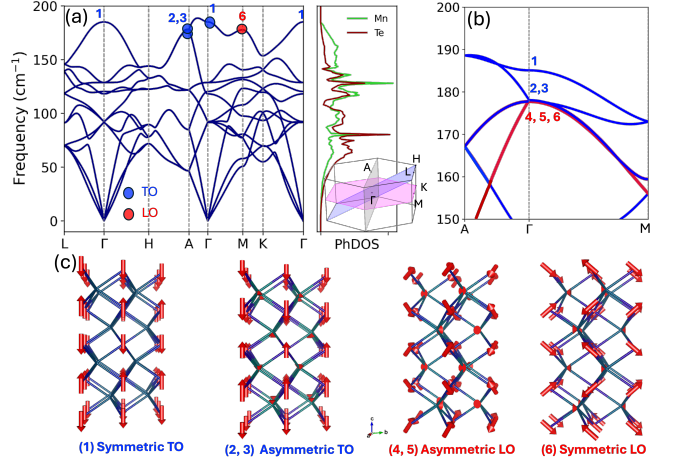


FIG. 1. (a) Phonon band structure and atom projected phonon density of states (PhDOS) of bulk MnTe without LR-correction. The high-symmetry points connecting the q-path is indicated in the Brillouin zone (inset). The blue and red points on the spectra at Γ , A and M points indicate three independent transverse optical (TO) and, one longitudinal optical (LO) vibrations observed in MnTe system. (b) The phonon band structure for the $2\times 2\times 2$ supercell of MnTe from which the bands are unfolded to represent the primitive unit cell band structure, as in Fig. (a). The blue and red bands are only shown for the high frequency TO and LO optical phonons, respectively. Once unfolded, the modes, 2-6 do not lie at Γ and are mapped to other high symmetry points of the primitive unit of MnTe. We mapped (2, 3) and (6) belong to A and M, respectively. The vibrations corresponding to these identified LO and TO modes are shown in panel (c).

tion. The two doubly degenerate modes (2, 3), at 178.05 cm^{-1} , correspond to asymmetric transverse optical (A-TO) vibrations. Here, symmetric and asymmetric refer to the displacement patterns of equivalent Mn and Te atoms within the supercell. The three degenerate longitudinal optical (LO) modes occur at 177.74 cm^{-1} and consist of two asymmetric LO (A-LO) and one symmetric LO (S-LO) modes. The displacement patterns of these modes are represented by vectors attached to the corresponding atomic sites in Fig. 1(c). When these supercell phonon modes are unfolded onto the primitive-unit-cell BZ, the S-TO mode remains at Γ , whereas the two A-TO modes are mapped onto the high-symmetry point A, as marked in Fig. 1(a). Among the three LO modes, only the S-LO mode appears at the M point, while the other two A-LO modes could not be discerned in the unfolded phonon bandstructure. Since, all these modes are lying in the region of interest (175 cm^{-1}), it is important to analyze these vibrations along with the zone-center one to provide a broader understanding of the high frequency vibraional response of MnTe.

The supercell convergence test is a necessary sanity check while calculating the phonon modes. Keeping

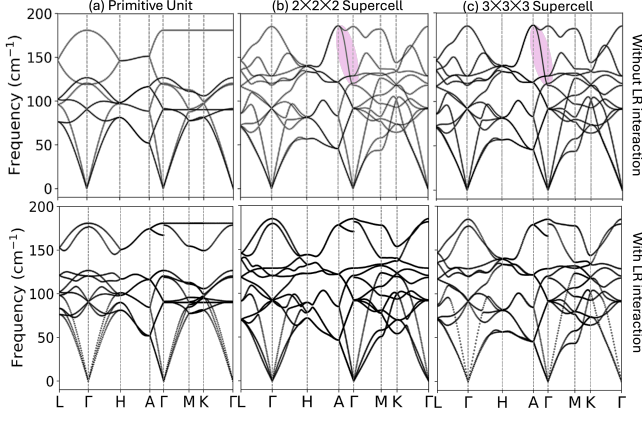


FIG. 2. Phonon band structures of MnTe with varying supercell size. The upper and lower panels show the band structures with and without LR-corrections respectively. The long-range dipole-dipole contribution to the dynamical matrix, resulted in improved convergence with a $2 \times 2 \times 2$ supercell, which is the minimum supercell size to retain all the operational symmetries for the space group.

this in mind, in the upper panel of Fig. 2 we compare the phonon band structure obtained from primitive unit cell, $2 \times 2 \times 2$ and $3 \times 3 \times 3$ supercells. Each of them show distinct differences, particularly involving the optical phonon modes. It is interesting to note that, the band structure shown by Wu *et al.* matches with the one obtained with $3 \times 3 \times 3$ supercell (Fig. 2(c))[8].

From the Fig. 2 upper panel, we also gather that the phonon bands around ~ 180 and ~ 90 cm^{-1} are sensitive to the supercell size. Specifically, along the path Γ -M-K- Γ , the flat band observed for the primitive unit becomes oscillatory as we increase the supercell. This is due to the fact that with increase in the supercell size, smaller phonon wavevectors (First BZ becomes even smaller) are sampled to describe the same vibrations. The other interesting observation is the presence of unusually steep phonon dispersion along Γ -A for $2 \times 2 \times 2$ and $3 \times 3 \times 3$ supercells. Such an abrupt dispersion near Γ can be linked with the finite size truncation of the vibration space. In fact, it is known that estimation of accurate phonon modes require very large supercells. Particularly, in the case of semiconductors and insulators, the incomplete electronic screening in the case of smaller supercell can lead to inaccurate estimation of the interatomic force constants which become a prominent factor in case of polar systems like NaCl, MgO, KCl etc.[17, 18]. In such systems, the finite size truncation can lead to Gibb's oscillation [16, 19]. As it is practically impossible to compute an infinitely large supercell for the calculations, alternatively, one can consider the non-analytic correction (NAC) using the Born effective charges and dielectric tensor proposed by Gonze *et al.*[19] to account for the long-range (LR) behaviour. Indeed in our system, we notice relatively large directional Born effective charges.

In the long-wavelength limit, the NAC contribution to

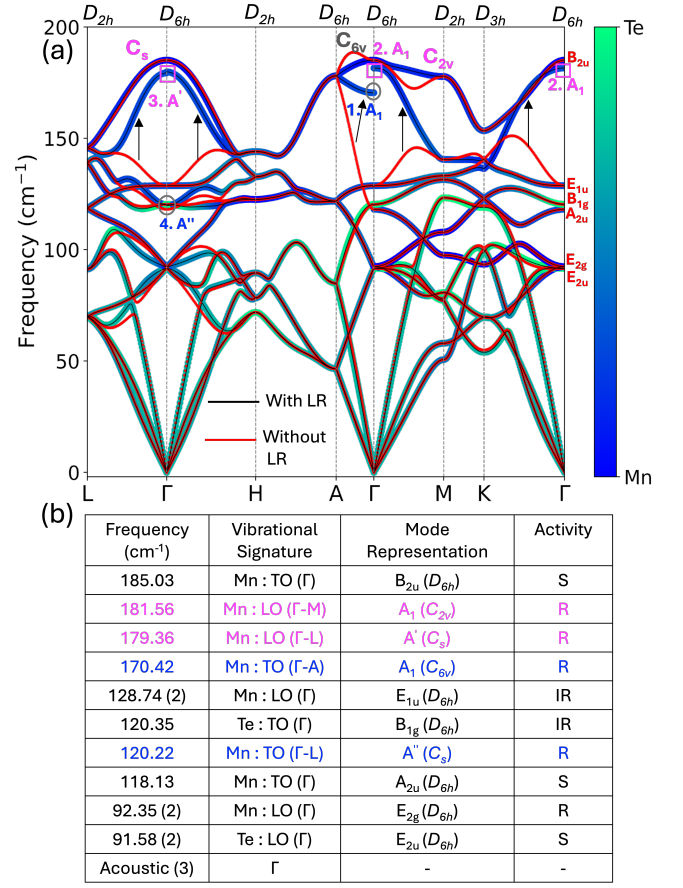


FIG. 3. (a) The detailed comparison of phonon spectra without LR interaction (red solid lines) and with LR interaction (color projected Mn and Te contributions). The inclusion of LR lifts the degeneracy in high frequency optical modes along certain symmetry paths. The direction dependent split of LO-TO modes are not well-defined at Γ , as they show abrupt discontinuity with the inclusion of LR. (b) The table listing the existing and new modes with their group theoretical representations. As observed, without LR interaction, phonon modes were either assigned to be silent or, infrared active. After the LR driven splitting it is found that, the new peaks that emerged around zone-center are all Raman active phonons (according to selection rules) around the frequency range of 120 -190 cm^{-1} .

the dynamical matrix can be written as[19]

$$D_{\kappa\alpha, \kappa'\beta}^{\text{NA}}(\mathbf{q} \rightarrow 0) = \frac{4\pi e^2}{\Omega} \frac{[\mathbf{q} \cdot \mathbf{Z}_{\kappa}^*]_{\alpha} [\mathbf{q} \cdot \mathbf{Z}_{\kappa'}^*]_{\beta}}{\sqrt{M_{\kappa} M_{\kappa'}} (\mathbf{q} \cdot \epsilon^{\infty} \cdot \mathbf{q})}, \quad (1)$$

where Ω is the unit-cell volume, M_{κ} and $M_{\kappa'}$ are the atomic masses, $\mathbf{Z}_{\kappa}^*$ and $\mathbf{Z}_{\kappa'}^*$ are the Born effective charge tensors, and ϵ^{∞} is the electronic dielectric tensor. The total dynamical matrix near the Γ point was therefore obtained by,

$$D(\mathbf{q}) = D^{\text{A}}(\mathbf{q}) + D^{\text{NA}}(\mathbf{q} \rightarrow 0), \quad (2)$$

where D^{A} represents the analytic part obtained from the short-range interatomic force constants.

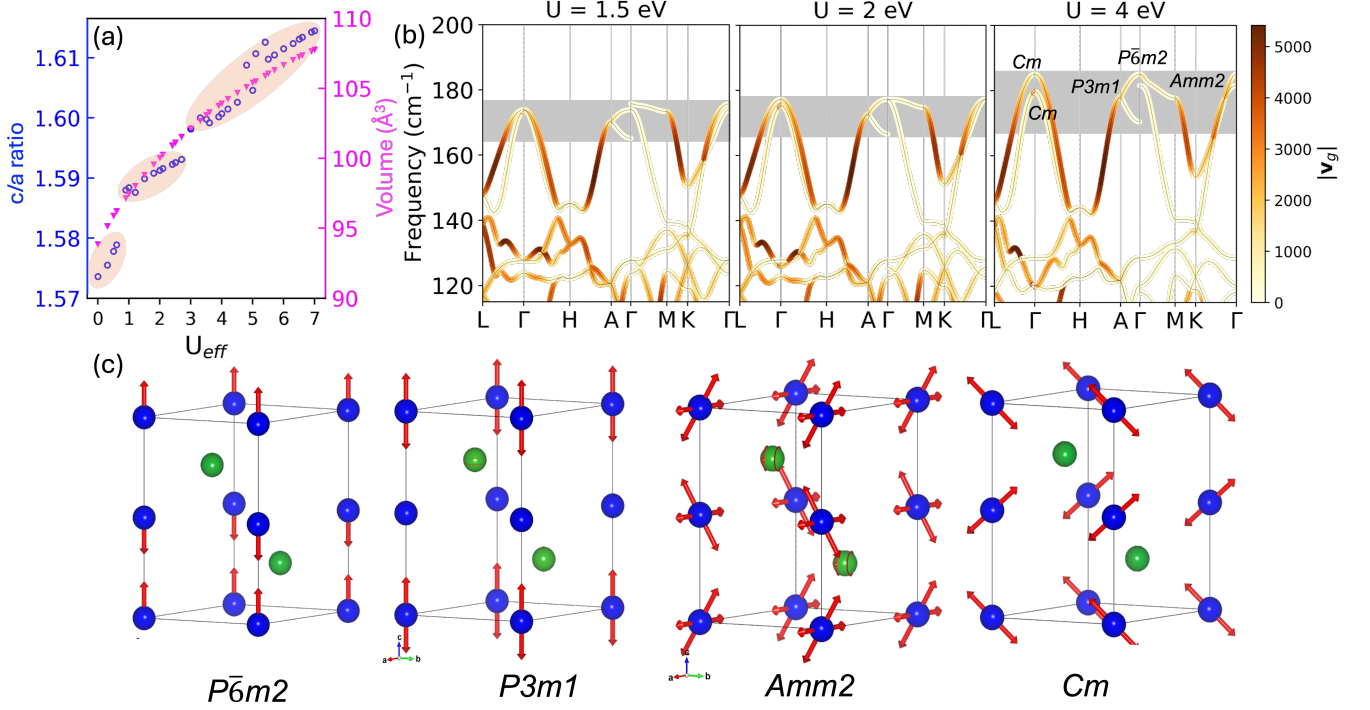


FIG. 4. (a) The c/a ratio and the primitive-cell volume as a function of the effective Hubbard parameter (U_{eff}). (b) Group-velocity (v_g) maps overlaid with the LR-corrected phonon band structures for $U_{eff}=1.5$ eV, 2 eV, and 4 eV. The results indicate the U-dependent behaviour of the high frequency optical phonon branches. (c) Schematic illustrations of the structural distortions generated by displacements along the corresponding LO and TO phonon eigenvectors of MnTe. The space groups of the resulting distorted structures are indicated.

By incorporating NAC, we have plotted the phonon band structure in the lower panel of Fig. 2, which not only implement significant corrections to the optical phonon modes but also shows a convergence with a $2 \times 2 \times 2$ supercell. To further elucidate the distinct features introduced by LR interaction, in Fig. 3, we represent the atom projected phonon band structure and compared them with the bands obtained without LR contribution (red solid lines). We would also like to add that, since, the corrected optical modes are not well-defined at Γ due to directional-dependence, we assigned their symmetry by comparing their characters with the appropriate irreducible representations along the corresponding high symmetry q-path in the BZ. The important changes by the LR correction, indicated by arrows in Fig. 3(a), are summarized here. (i) The steep TO phonon dispersion rising from 118 cm^{-1} at Γ - A disappears and instead a new TO branch emerge from the same point at A and reaches 170.4 cm^{-1} at Γ . This TO mode, encircled in grey, follows the C_{6v} symmetry of the Γ -A path and has A_1 representation. (ii) The LO branch starting from 140 cm^{-1} at M and reaching at 128.8 cm^{-1} Γ is now lifted at the same point to 181.6 cm^{-1} . A similar one is observed while going from K- Γ . The mode is marked by a pink open rectangle and it also has A_1 character based on the C_{2v} symmetry along the Γ -M (K) path. (iii) Focusing

on the path along L- Γ -H, we find that a LO branch in the range 128 cm^{-1} changes the curvature from downward to upward, as indicated by arrows, and reaches $179 \pm \delta \text{ cm}^{-1}$ with a discontinuity at Γ . This corrected LO mode (pink rectangle) follows the C_s symmetry for Γ -L (H) path and has A' character. (iv) The one of the TO mode at Γ , marked with a grey circle, shifts from 118 to 120 cm^{-1} . This mode is defined along the same Γ -L direction and has A'' character.

Fig. 3(b) lists the existing and new modes (after LR correction) with their group theoretical representations and their optical activity. Interestingly, we find that all the four new modes, A_1 (TO), A_1 (LO), A' (LO), A'' (TO), denoted in Fig. 3(a), are Raman active according to the selection rules of their respective path symmetries. Therefore, the LR interaction corrected phonon band structure along with the newly resolved vibrational modes provide a basis for examining the origin and robustness of the Raman response measured in bulk MnTe.

The strong correlation has a profound effect on the electronic structure of MnTe[20, 21]. With change in the electronic structure the mean-field in which the lattice vibration occurs also changes. Therefore, it is prudent to examine if the strong correlation measured through U , also has an impact on the vibrational properties of MnTe. In this context, we carried out structural relaxation as a

function of U by varying it from 0 to 7 eV. The resulted lattice volume and c/a ratio are presented in Fig. 4(a). Both the lattice volume and c/a increase progressively with increasing U , indicating an evolution of the equilibrium structure. We also note that, with increase in c/a , the system achieves a saturated magnetic state consistent with previous reports[20, 22].

Subsequently, LR interaction included phonon band structure for each value of U , using the corresponding relaxed lattice parameters are calculated and the results are shown in Fig. 4(b) for three U values (1.5 eV, 2 eV, and 4 eV). From the figure we gather that, there are three major changes around the zone-center with the change in U . First, the optical modes gradually shift toward higher frequencies with increasing U . Second, by observing the bands along Γ -A- Γ -M, we find that the relative LO and TO ordering at the highest branch changes with U , resulting in a LO-TO crossover as U increases (marked by a shaded region). Third, a distinct splitting occurs between Mn and Te dominated vibrational branches around 90 and 120 cm^{-1} (not shown here). These changes implies that the vibrational response of MnTe is strongly coupled to the U -dependent structural changes. In particular, the crossover of modes around the zone-center is important to understand the distortion mediated changes in optical phonons in pristine MnTe. Based on these distinct structural and vibrational trends, the U dependent behavior can be roughly divided into three characteristic regimes, namely, $0 \leq U \leq 1.8$ eV; $2 \leq U \leq 2.8$ eV; and $3.0 \leq U \leq 7.0$ eV, as indicated in Fig. 4(a). This figure also implies that, depending on the value of U used in the theoretical calculations, the predicted positions of the Raman peaks will vary.

As shown in Fig. 4(b), the dispersion becomes gradually steeper with increasing U , specially for the highest optical branch. To further characterize the directional behaviour of these high frequency vibrations, we examine the phonon group velocity, v_g , along the three independent high-symmetry directions, Γ -A, Γ -M, and Γ -L, respectively. Since $v_g=0$, exactly at the zone-center for the optical modes, we focus on the dispersion gradient in the vicinity of Γ for the two optical branches. For the first and second regimes ($0 \leq U \leq 1.8$ eV; $2 \leq U \leq 2.8$ eV), the highest LO, TO modes both exhibit negligible v_g along Γ -M and, Γ -A, respectively. Above $U \geq 4.0$ eV, however, the dispersion near Γ becomes steeper, with the v_g for TO along Γ -M (25.51 km/s) becoming approximately five times larger than that along Γ -A (5.97 km/s). In contrast, the Γ -L direction consistently exhibits the largest v_g for both the TO and LO modes across all U range. In fact, we find that the v_g of TO (255.73 km/s) and, LO (2613.92 km/s) modes along Γ -L are nearly two and three orders of magnitude larger, respectively, than those along Γ -M. The directional dependence of v_g , reveals that, the progression of the zone-center modes associated with specific distortions can be a prominent pathway to understand the vibrational fingerprint of MnTe.

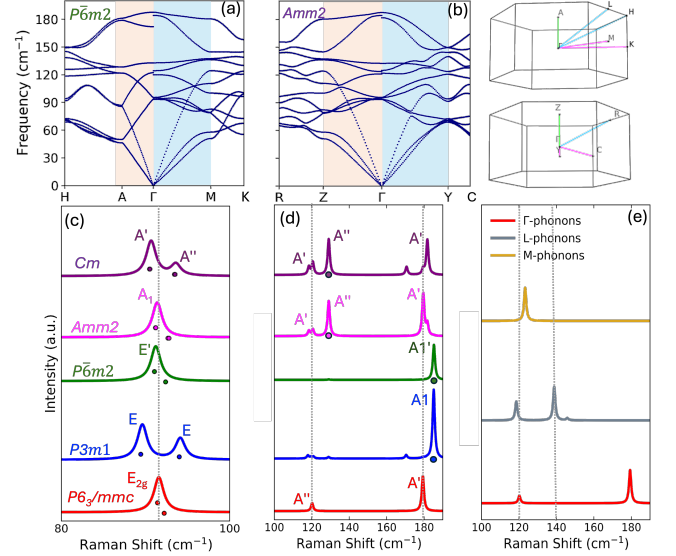


FIG. 5. (a, b) Phonon band structures for two distinct type of TO and LO distortion of MnTe, respectively. The atoms are displaced along the identified modes to the tune of mass-normalized 0.1 Å. Their corresponding BZs are shown on the right panel. LR- corrected Raman spectra of pristine MnTe with all the plausible distorted analogues for two separate frequency range of (c) 80 - 100 cm^{-1} and (d) 100 - 190 cm^{-1} . Without LR interaction, the obtained Raman peak positions are also included as colored dots in the spectra. (e) The off-center phonon contribution to the Raman spectra emerging out of the M and L - centered phonons and compared to that of the zone centered phonons.

Now, we examine the structural alterations arising from the high frequency vibrations. This is achieved by “freezing” the phonon eigenvectors of the six modes (three LO and three TO) and analyzing the resulting symmetry lowered structures. Each mode produces a unique distortion in MnTe and lowers the parent space group symmetry as indicated in, $P6_3/mmc$, to a corresponding maximal subgroup symmetry, as indicated in Fig. 4(c). The distortion directions can be visualized from the lower panel of Fig. 4. The freezing of the S-TO mode yields a $P\bar{6}m2$ (No. 187) structure, whereas the A-TO mode produces a $P3m1$ (No. 156) structure. The A-LO and S-LO modes, in turn, lead to even lower symmetric $Amm2$ (No. 38) and Cm (No. 6) structures, respectively.

The LR corrected phonon band structures of distorted MnTe are shown in Fig. 5(a, b) for two representative LO and TO distortions. The $P\bar{6}m2$ distortion produces only minor changes to the phonon bands of the pristine MnTe. On the other hand, as highlighted, the $Amm2$ distorted structure, presented in a different BZ (Fig. 5(b)), exhibits contrasting phonon dispersion along the transverse Γ -Z and longitudinal Γ -Y directions, when compared to the dispersion along the transverse Γ -A and longitudinal Γ -M for the pristine MnTe. As observed from Fig. 5(b), $Amm2$ type distortion reduces the directional disparity

of the modes around zone-center. The band structure corresponding to $P\bar{6}m2$ and $Amm2$ are nearly identical to $P3m1$ and Cm , respectively.

To comprehend the overall changes in the vibrational landscape of MnTe, discussed so far, we further calculate the first-order Raman spectra of pristine and distorted MnTe by taking into account NAC. The zone-center Raman spectra are presented in Fig. 5(c) and (d) for two separate frequency ranges of 80 - 100 cm^{-1} and 100 - 190 cm^{-1} , respectively. The position of the peaks without LR interaction are also indicated by colored dots. We note that, several experimental studies[11, 12] have reported four distinct Raman features below 200 cm^{-1} , centered around 90 cm^{-1} , 122 cm^{-1} , 140 cm^{-1} , and 175 cm^{-1} . Although theoretical calculations ignoring the LR dipole-dipole interactions have correctly reproduced the 90 cm^{-1} peak[8, 9], they have failed to do so for the remaining three. Therefore, the microscopic origin of the other three peaks has remained far from resolved. Various mechanisms, including distortion-mediated and plasmon-related effects, have been proposed to account for the higher frequency Raman features in MnTe (see the first section). In this study, by explicitly incorporating the LR dipole-dipole interaction and resolving the associated LO-TO splitting, we reproduce all four experimentally observed Raman peaks under the appropriate structural and vibrational conditions.

The prominent Raman peak near 90 cm^{-1} in Fig. 5(c) corresponds to the robust E_{2g} type vibration of pristine MnTe. This mode remains the most intense Raman peak irrespective of the structure being pristine or distorted. However, the distortions $P\bar{6}m2$ and Cm cause a small split of this mode by $\sim 4 \text{ cm}^{-1}$, while preserving its characteristic vibrational pattern. Ukyur *et al.* in their experimental studies have reported the presence of close-spaced double peaks corresponding to this E_{2g} mode[9].

As observed in Fig. 5(d), for pristine MnTe, the peak near 120 cm^{-1} originates from the branch along Γ -L direction, revealed by the LR correction [see Fig. 3(a)]. Without LR correction, this mode has the A_{2u} symmetry and is Raman inactive. The LR interaction shifts this branch (as discussed earlier through Fig. 3) and makes this otherwise hidden mode Raman active. Its response also depends strongly on the distortion, such as, the TO-type distortions ($P\bar{6}m2$, $P3m1$) suppress the 120 cm^{-1} feature, whereas the LO-type distortions ($Amm2$, Cm) intensifies it. LO type distortions also give rise to another peak at 129 cm^{-1} which was also silent (E_{1u}) in pristine MnTe.

The LR dipole-dipole interactions introduce a new Raman peak in pristine MnTe that appears at 179 cm^{-1} (see Fig. 5(d)). We traced its origin to the splitting of the S-LO phonon along the Γ -L direction (see Fig.

3(a)) and we attribute it as one of the reasons for experimentally observing a peak around 175 cm^{-1} . The same feature is retained in the both LO-distorted structures, but disappears for the TO-type distortions. Instead, the TO-distorted structures develop another A_1 -type Raman mode near 185 cm^{-1} both with and without LR correction. This is consistent with the discussions made by Wu *et al.*, where the authors have pinned down the highest TO mode to be Raman active under this specific (S-TO distortion) as the symmetry of this mode changes from B_{2u} to A'_1 [8]. Furthermore, S-LO distortion corresponding to Cm symmetry shows a Raman active mode at 170 cm^{-1} (see the upper panel Fig. 5(d)).

Next, we discuss the experimental Raman peak $\sim 140 \text{ cm}^{-1}$. Previous reports have attributed it to the existence of zone off-center Raman modes [8, 9]. From our calculated Raman spectra, we indeed found a Raman peak at 140 cm^{-1} (see Fig. 5(e)) and they originate from zone boundary L-phonons. In addition, we see two new peaks around 120 cm^{-1} arising from L and M phonons. Therefore, 120 cm^{-1} peak is collectively formed by zone-center and zone-boundary phonons. Since this feature at $\sim 140 \text{ cm}^{-1}$ does not arise from zone-center phonons, but instead comes to existence from off-center contributions at L and M, the experimental detection of this particular peak may indicate the activation of phonons through higher-order Raman processes[23, 24].

To summarize and conclude, we provide theoretical insights into the experimental landscape in Raman response of MnTe. The LR dipole-dipole interaction is the key to accurately calculate the phonon band structure of this altermagnetic insulator and it is due to the fact that it exhibits large directional-dependent Born effective charges. The LR corrections reveal the Raman active modes around 120 cm^{-1} , 140 cm^{-1} , and 179 cm^{-1} which are in-addition to the existing 90 cm^{-1} peak that is reproduced even without LR correction. Further analysis shows that the 140 cm^{-1} peak originates from the zone-boundary phonons, while the 120 cm^{-1} peak has contribution from both zone-center and zone boundary phonons. The highly debated Raman peak around 175 cm^{-1} seems to be of many origins, including the LR effects in pristine MnTe, LO distortions, and the presence of impurity MnTe_2 phases as seen in many experimental studies. We further emphasize that the more probable in-plane distortions induced by S-LO phonons with high group-velocity (associate with Cm type distortion) can produce intense Raman response at ~ 120 and 179 cm^{-1} . Furthermore, the out-of-plane distortions induced by A-TO phonons generate a single Raman peak that lies well above 179 cm^{-1} . Our prescribed distortions driven Raman response will encourage further experimental research in this direction.

[1] L. Šmejkal, J. Sinova, and T. Jungwirth, Physical Review X **12**, 031042 (2022).

[2] R. Sarkar, S. K. Negi, A. Das, A. Mandal, P. Bhardwaj,

- S. Guin, A. Das, N. Shyaga, L. Nanda, B. Nanda, *et al.*, arXiv preprint arXiv:2605.25953 (2026).
- [3] S. Li, J. Wu, B. Liang, L. Liu, W. Zhang, N. Wazir, J. Zhou, Y. Liu, Y. Nie, Y. Hao, *et al.*, Chemistry of Materials **34**, 873 (2022).
- [4] S. Lee, S. Lee, S. Jung, J. Jung, D. Kim, Y. Lee, B. Seok, J. Kim, B. G. Park, L. Šmejkal, *et al.*, Physical review letters **132**, 036702 (2024).
- [5] M. Chilcote, A. R. Mazza, Q. Lu, I. Gray, Q. Tian, Q. Deng, D. Moseley, A.-H. Chen, J. Lapano, J. S. Gardner, *et al.*, Advanced Functional Materials **34**, 2405829 (2024).
- [6] K. Kluczyk, K. Gas, M. Grzybowski, P. Skupiński, M. Borysiewicz, T. Faş, J. Suffczyński, J. Domagala, K. Graszka, A. Mycielski, *et al.*, Physical Review B **110**, 155201 (2024).
- [7] J. Zhang, Q. Lian, Z. Pan, W. Bai, J. Yang, Y. Zhang, X. Tang, and J. Chu, Journal of Raman Spectroscopy **51**, 1383 (2020).
- [8] A. Wu, D. Cheng, J.-Y. Chung, X. Wang, M. Zeng, C. Liu, S. Gradečak-Garaj, and X. Li, Physical Review B **114**, 074436 (2026).
- [9] E. Uykur, M. V. Gonçalves-Faria, S. Rößler, V. A. Ginga, M. Schmidt, S. Winnerl, M. Helm, and A. A. Tsirlin, Physical Review B **113**, 224107 (2026).
- [10] W. Szuszkiewicz, J. Mohrange, M. Jouanne, M. Kanehisa, R. Świrkowicz, E. Dynowska, E. Janik, T. Wojtowicz, and J. Kossut, Acta Physica Polonica A **92**, 1025 (1997).
- [11] N. Azam, S. M. Shahed, L. T. Schmidt, S. Bey, O. Yastrubchak, M. F. Munoz, R. Torsi, T. H. Y. Pham, D. Balasundaram, R. B. Regami, *et al.*, arXiv preprint arXiv:2609.19467 (2026).
- [12] P. Bhardwaj, R. Sarkar, N. Shyaga, S. K. Negi, K. Biswas, S. Jana, P. Nukala, B. R. K. Nanda, and D. Suri, Nature of the 175-180 cm^{-1} raman feature in altermagnetic α -mnTe (2026), arXiv:2609.39108 [cond-mat.mtrl-sci].
- [13] W. Szuszkiewicz, M. Jouanne, J.-F. Morhange, M. Kanehisa, E. Dynowska, K. Gas, E. Janik, G. Karczewski, R. Kuna, and T. Wojtowicz, physica status solidi (b) **251**, 1133 (2014).
- [14] T. Palasyuk, D. Kurzydłowski, A. S. Wadge, R. Bacewicz, M. H. Dalsaniya, M. Wierzbicki, D. Jas-trzebski, A. Wiśniewski, W. Szuszkiewicz, I. Antoniazzi, *et al.*, Journal of Alloys and Compounds , 189349 (2026).
- [15] S. L. Dudarev, G. A. Botton, S. Y. Savrasov, C. J. Humphreys, and A. P. Sutton, Physical Review B **57**, 1505 (1998).
- [16] A. Togo, L. Chaput, T. Tadano, and I. Tanaka, J. Phys. Condens. Matter **35**, 353001 (2023).
- [17] A. Togo, H. Hayashi, T. Tadano, S. Tsutsui, and I. Tanaka, Journal of Physics: Condensed Matter **34**, 365401 (2022).
- [18] G. Coiana, J. Lischner, and P. Tangney, Physical Review B **109**, 014310 (2024).
- [19] X. Gonze and C. Lee, Physical Review B **55**, 10355 (1997).
- [20] E. D. Chernov and A. V. Lukoyanov, Materials **18**, 2637 (2025).
- [21] B. Kang, A. Janotti, D. Q. Ho, M.-H. Kim, C. H. Park, S. Choi, M. R. Pederson, and E. Kim, arXiv preprint arXiv:2605.14248 (2026).
- [22] E. P. Carlisle, G. Yumnam, S. Calder, B. Haberl, J.-X. Xiong, M. A. McGuire, A. Zunger, R. P. Hermann, and B. A. Frandsen, Physical Review B **112**, 014450 (2025).
- [23] H. Padmanabhan, M. Poore, P. K. Kim, N. Z. Koocher, V. A. Stoica, D. Puggioni, H. Wang, X. Shen, A. H. Reid, M. Gu, *et al.*, Nature communications **13**, 1929 (2022).
- [24] G. Güntherodt, W. Bauhofer, and G. Benedek, Physical Review Letters **43**, 1427 (1979).